

Digital Electronic Systems Design Project Report

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Timing Analysis

The implemented design demonstrates the timing analysis yields compliant results for both setup and hold constraints at the target operating frequency of 150.89286 MHz.

Regarding the **setup time**, the most critical path (Path 21) is located within the `reverb_0` sub-block. While the design requirement demands a 150.89286 MHz clock, this implementation achieves a positive slack of 0.299 ns. This safety margin allows the circuit to potentially operate at a maximum frequency of approximately 158 MHz, derived as $f_{\max} = 1 / (6.627 \text{ ns} - 0.299 \text{ ns}) \approx 158.0 \text{ MHz}$.

Concerning the **hold time**, the most critical path (Path 31) is located within the `axis_fifo_0` sub-block. The required hold time constraint is 0 ns, and the implementation guarantees a safe positive hold slack of 0.024 ns, ensuring that no race conditions occur.

The detailed parameters resulting from the implementation are summarized in Table 1 for setup time and in Table 2 for hold time below, related to the above mentioned most critical paths.

Table 1: Timing Summary: Setup Time (Path 21)

Parameter	Value	Unit
Slack	0.299	ns
Requirement	6.627	ns
Data Path Delay	2.378	ns
<i>Logic Delay</i>	0.742	ns
<i>Route Delay</i>	1.636	ns
Clock Path Skew	0.055	ns
Clock Uncertainty	0.079	ns

Table 2: Timing Summary: Hold Time (Path 31)

Parameter	Value	Unit
Slack	0.024	ns
Requirement	0.000	ns
Data Path Delay	0.347	ns
<i>Logic Delay</i>	0.141	ns
<i>Route Delay</i>	0.206	ns
Clock Path Skew	0.013	ns

Power Analysis

At the operating frequency of 150.89286 MHz, the on-chip power consumption of the implemented design totals 0.334 W, broken down into a **dynamic power** component of 0.261 W (78%) and a **static power** (device leakage) component of 0.074 W (22%). The dominant contributor to dynamic power is the MMCM clock manager at 0.126 W (48% of dynamic), followed by BRAM at 0.078 W (30%) and I/O at 0.017 W (6%). The junction temperature of the package is estimated at 26.7 °C, with a thermal margin of 58.3 °C and an effective thermal resistance θ_{JA} of 5.0 °C/W, indicating comfortable thermal headroom under the given operating conditions. The confidence level of the power estimate is reported as Low, as no switching activity data from simulation was provided, and the analysis relies on vectorless estimation. The detailed power figures are reported in Table 3 and Table 4.

Table 3: Power Analysis Summary.

Parameter	Value	Unit
Total On-Chip Power	0.334	W
Design Power Budget	N/A	–
Power Budget Margin	N/A	–
Junction Temperature	26.7	°C
Thermal Margin	58.3	°C
Effective θ_{JA}	5.0	°C/W
Power to Off-Chip Devices	0	W
Confidence Level	Low	–

Table 4: On-Chip Power Breakdown.

Component	Power	Unit	Ratio (%)
Dynamic	0.261	W	78%
<i>MMCM</i>	0.126	W	48%
<i>BRAM</i>	0.078	W	30%
<i>I/O</i>	0.017	W	6%
<i>Clocks</i>	0.018	W	7%
<i>Signals</i>	0.012	W	5%
<i>Logic</i>	0.009	W	3%
<i>DSP</i>	0.002	W	1%
Device Static	0.074	W	22%